

1.Create a comparison table listing RAID 0, RAID 1, RAID 5, and RAID 10, including columns for minimum number of disks, data redundancy, speed, and a real-world use case for each (such as Instagram photo storage, YouTube video servers, etc.).

RAID Level	Minimum Disks	Data Redundancy	Speed	Real-World Use Case
RAID 0	2	✗ No redundancy	🚀 Very High	Video editing / temporary YouTube video processing
RAID 1	2	✓ Full mirroring	⚡ Good read speed	Important office files / small database
RAID 5	3	✓ Can tolerate 1 disk failure	⚡ Good	Instagram photo storage / file servers
RAID 10	4	✓ Mirroring + striping	🚀 Very High	YouTube video servers / high- traffic databases

2.Explain with your own example how RAID 1 provides data redundancy, using a scenario like storing WhatsApp chat backups on two disks, and describe what happens if one disk fails.

Ans:-

Suppose a server stores **WhatsApp chat backups** using **RAID 1** with two disks:

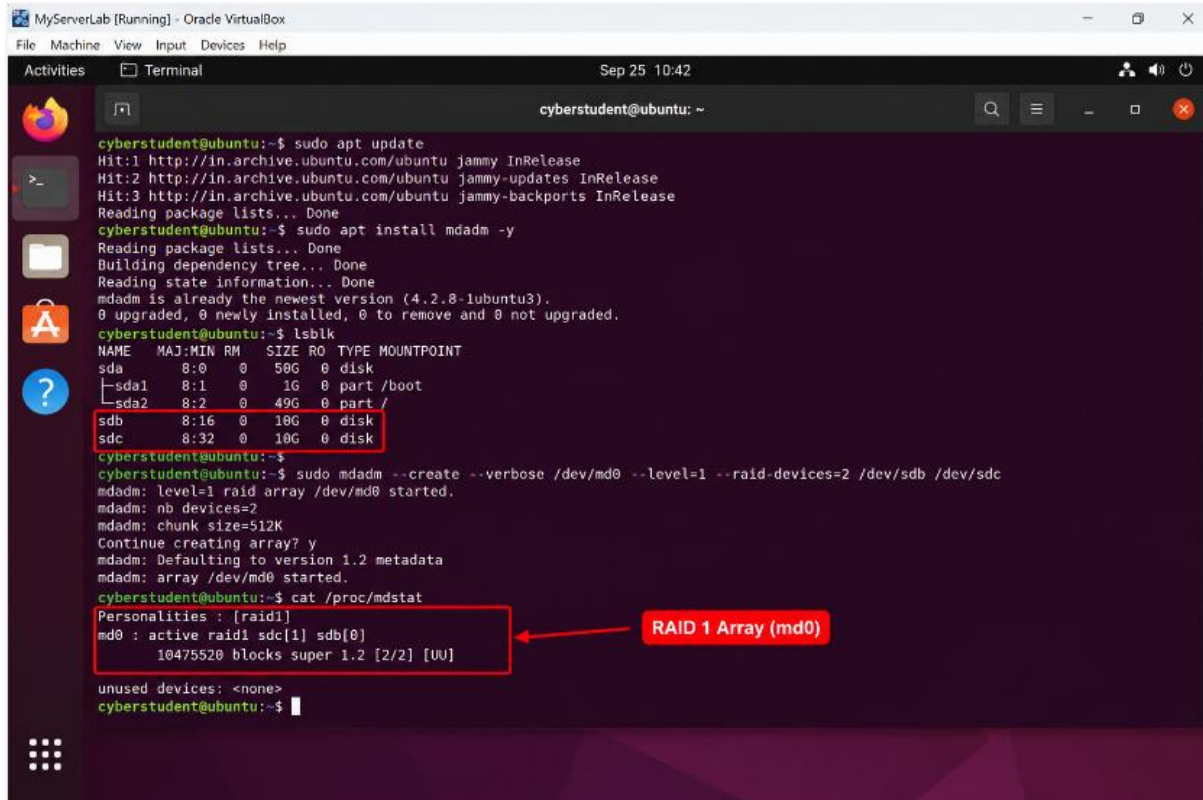
- **Disk 1:** Stores the WhatsApp chat backup.
- **Disk 2:** Stores an **exact copy (mirror)** of the same backup.

For example, if the chat backup is **100 GB**, both Disk 1 and Disk 2 contain the same 100 GB data.

If Disk 1 fails:

- The server continues working using **Disk 2**, because Disk 2 has an identical copy of the data. No data is lost.
- After replacing the failed Disk 1, the RAID system can copy the data from Disk 2 to the new disk and restore the mirror.
- **Conclusion:** RAID 1 provides **data redundancy through mirroring**, so the system can continue operating even if one disk fails.

3. Simulate a RAID 1 setup using VirtualBox and Ubuntu: create two virtual hard disks, use the 'mdadm' tool to configure them in RAID 1, and take a screenshot of the output from the command 'cat /proc/mdstat' showing the RAID array. **Hint:** Search for 'mdadm RAID 1 Ubuntu VirtualBox' for step-by-step guides.



The screenshot shows a terminal window titled 'MyServerLab [Running] - Oracle VirtualBox' with a menu bar (File, Machine, View, Input, Devices, Help) and a status bar (Sep 25 10:42). The terminal is running as 'cyberstudent@ubuntu: ~'. The user performs the following commands and receives the following output:

```
cyberstudent@ubuntu:~$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
cyberstudent@ubuntu:~$ sudo apt install mdadm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mdadm is already the newest version (4.2.8-1ubuntu3).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cyberstudent@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
sda   8:0    0   50G  0 disk
├─sda1 8:1    0    1G  0 part /boot
└─sda2 8:2    0   49G  0 part /
sdb   8:16   0   10G  0 disk
sdc   8:32   0   10G  0 disk
cyberstudent@ubuntu:~$ sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc
mdadm: level=1 raid array /dev/md0 started.
mdadm: nb devices=2
mdadm: chunk size=512K
Continue creating array? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
cyberstudent@ubuntu:~$ cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdc[1] sdb[0]
      10475520 blocks super 1.2 [2/2] [UU]

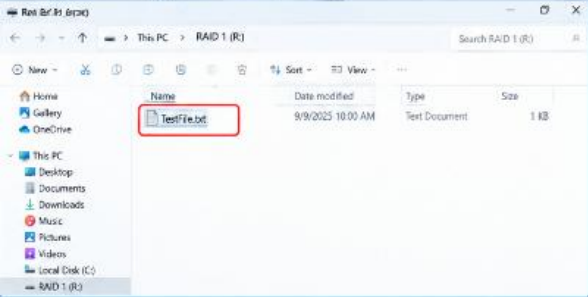
unused devices: <none>
cyberstudent@ubuntu:~$
```

Red boxes highlight the `lsblk` output for `sdb` and `sdc`, and the `cat /proc/mdstat` output. A red arrow points from a red box labeled "RAID 1 Array (md0)" to the `md0` line in the `cat /proc/mdstat` output.

4.Delete a file from your RAID 1 array, then power off one of the virtual disks (simulate a disk failure), and check if the file is still accessible from the RAID array.

Hint: This tests if RAID 1 is really mirroring your data.

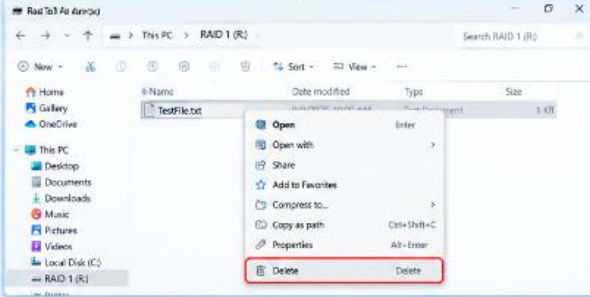
1. My RAID 1 array (before deleting the file)



File Explorer window showing the RAID 1 array. The file 'TestFile.txt' is highlighted in the list.

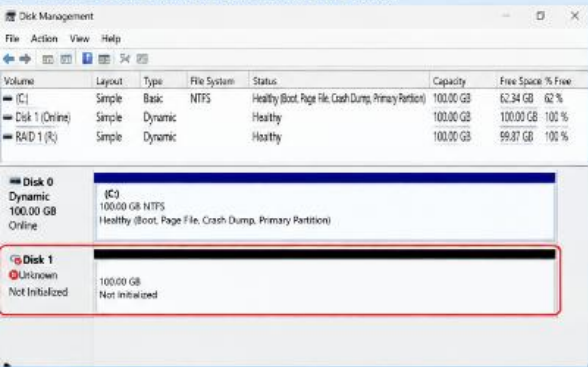
Name	Date modified	Type	Size
TestFile.txt	9/9/2025 10:00 AM	Text Document	1 KB

2. Delete the file from the RAID 1 array



File Explorer window showing the context menu for 'TestFile.txt'. The 'Delete' option is highlighted.

3. Power off one virtual disk (simulate a disk failure)

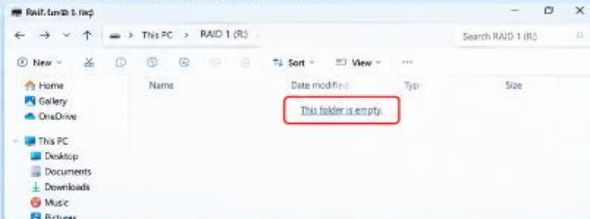


Disk Management window showing the status of the disks. Disk 1 is highlighted in red, indicating it is 'Unknown' and 'Not Initialized'.

Volume	Layout	Type	File System	Status	Capacity	Free Space	% Free
(C:)	Simple	Basic	NTFS	Healthy (Boot, Page File, Crash Dump, Primary Partition)	100.00 GB	62.34 GB	62 %
Disk 1 (Online)	Simple	Dynamic		Healthy	100.00 GB	100.00 GB	100 %
RAID 1 (R:)	Simple	Dynamic		Healthy	100.00 GB	99.97 GB	100 %

Disk	Dynamic	Capacity	File System	Status
Disk 0	Dynamic	100.00 GB	NTFS	Online
Disk 1	Unknown	100.00 GB		Not Initialized

4. Check the file from the RAID 1 array



File Explorer window showing the RAID 1 array. The message 'This folder is empty' is displayed, indicating the file has been deleted.

5. Result



Microsoft Windows error message: 'The item 'TestFile.txt' can't be found. It is no longer located in RAID 1 (R:).'.

5. Use ChatGPT to ask: 'What are the main differences between RAID 5 and RAID 10 for a cloud storage service like Google Drive?' Summarize the answer in 3 bullet points in your own words.

Ans:-

- **RAID 5:** Uses parity to protect data and needs at least 3 disks. It provides good storage efficiency but slower write performance.
- **RAID 10:** Combines mirroring and striping, needs at least 4 disks, and gives faster read/write performance with better fault tolerance.
- **For a cloud service:** RAID 5 uses storage space more efficiently, while RAID 10 offers higher performance and redundancy but requires more storage capacity.